Explaining Facial Emotion Recognition: Understanding Key Facial Features through Multi-Method XAI Analysis

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Abstract

Fatial Emotion Recognition(FER) models based on deep convolutional neural networks have achieved high classification performance, yet their decision-making processes remain largely opaque. This lack of transparency raises concerns regarding reliability, bias, and alignment with human perceptual mechanisms.

This project investigates how a CNN-based FER model trained on the FER2013 dataset makes emotion predictions by applying multiple Explainable AI(XAI) techniques, including Grad-CAM, Integrated Gradients, LIME, and RISE. Rather than relying on a single interpretability method, this study adopts a multi-method perspective to compare explanation patterns across techniques. In addition to qualitative visualization, explanation faithfulness is quantitatively evaluated using insertion and deletion metrics. The results demonstrate that while different XAI methods highlight distinct facial regions, consistent attention patterns emerge around psychologically meaningful areas such as the eyes and mouth. This analysis contributes to a deeper understanding of FER model behavior and supports the role of XAI in building trustworthy affective computing systems.

1. Introduction

Facial Emotion Recognition (FER) plays a critical role in affective computing, enabling machines to infer human emotional states from visual facial expressions. FER technologies are widely applied in areas such as human–computer interaction, mental health assessment, social robotics, and intelligent surveillance systems. With the advancement of deep learning, convolutional neural networks (CNNs) have become the dominant approach for FER due to their ability to automatically learn complex visual representations.

Despite their strong predictive capabilities, CNN-based FER models are often criticized for their lack of interpretability. These models operate as black boxes, offering little insight into why a particular facial expression is classified as a specific emotion. This lack of transparency can lead to mistrust, especially in socially sensitive applications, and raises questions about whether models rely on meaningful facial cues or spurious correlations.

Explainable Artificial Intelligence (XAI) seeks to address this limitation by providing tools that reveal how models arrive at their predictions. In the context of FER, XAI methods offer an opportunity to examine whether model attention aligns with established psychological theories of human emotion perception. This project focuses on applying and comparing multiple XAI techniques to analyze a CNN-based FER model, with the goal of understanding both where the model looks and how reliably these explanations reflect the model's true decision process.

1.1. Related Work

From a psychological perspective, studies on human facial expression recognition suggest that observers primarily rely on the eye and mouth regions when interpreting emotions. [1] Prior work has shown that different emotions emphasize different facial features, motivating comparisons between human attention patterns and model explanations. This project builds on both XAI and psychological literature by examining whether FER models attend to similar facial cues and how consistently different XAI methods reveal such patterns.

2. Method

2.1. Dataset

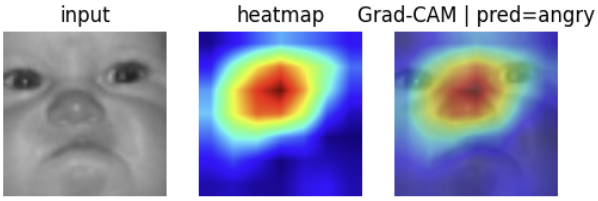
The FER2013 dataset [2], introduced in the 2013 ICML Representation Learning Challenge, contains 35,887 grayscale facial images (48×48 pixels) categorized into seven emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral. The images were collected from the internet using the Google Image API and include natural variations in lighting, pose, and expression intensity, making the dataset both challenging and realistic. In this project, the dataset is used to train and evaluate a CNN-based model,

and is preprocessed through standard resizing, normalization, and augmentation techniques to improve generalization and support subsequent XAI-based analysis.

2.2. XAI Techniques

2.2.1 Grad-CAM

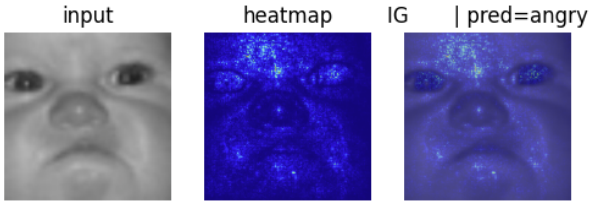
The Grad-CAM (Gradient-weighted Class Activation Mapping) code implemented in this project aims to identify which spatial regions of a facial image most strongly influence the model's prediction for a specific emotion class. Concretely, the code computes Grad-CAM on the last convolutional layer of the CNN, where spatial information is still preserved while semantic abstraction is high. By backpropagating the gradients of the predicted class score (e.g., angry) with respect to the feature maps of this layer, the implementation captures how strongly each spatial location contributes to increasing the model's confidence for that class.



Through this process, the Grad-CAM code operationalizes the question: "Where is the model looking in the face to justify its prediction?"

2.2.2 Integrated Gradients

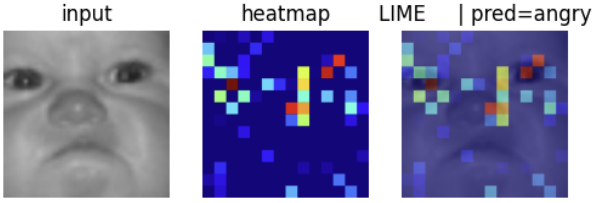
The Integrated Gradients (IG) implementation seeeks to quantify how individual input pixels contribute to the model's prediction for a specific emotion class, in contrast to the region-level explanations provided by Grad-CAM. In the implemented code, IG computes the integral of gradients along a straight-line path from a baseline image (an all-zero black image) to the original input image. By accumulating gradients across multiple interpolation steps, the method estimates the total contribution of each pixel to the final prediction score. The resulting attribution map is aggregated across channels, normalized, and visualized both as a standalone heatmap and as an overlay on the original input image.



Through this process, the IG code operationalizes the question: "Which exact pixels in the face most strongly influence the model's prediction?"

2.2.3 LIME

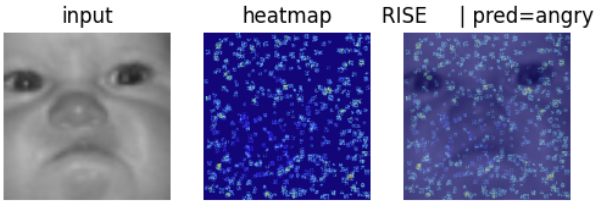
LIME (Local Interpretable Model-agnostic Explanations) aims to identify a compact set of localized facial regions that are most influential for a single prediction, focusing on local decision behavior rather than global model structure. Unlike gradient-based methods, LIME treats the FER model as a black box and approximates its behavior around a specific input by fitting a locally linear surrogate model. The facial image is divided into a fixed grid, and each grid cell is treated as an interpretable feature. Random perturbations are generated by masking subsets of grid cells, and the model's prediction scores for these perturbed samples are used to estimate feature importance. The resulting weights are aggregated and visualized as a patch-level heatmap.



Through this process, the LIME code operationalizes the question: "Which small, localized facial patches are sufficient to explain this specific prediction?"

2.2.4 RISE

RISE (Randomized Input Sampling for Explanation) aims to estimate the causal importance of input regions by directly measuring how random occlusions affect the model's prediction. Unlike gradient-based methods, RISE does not rely on internal model gradients or feature maps. Instead, it treats the model as a black box and evaluates explanation faithfulness through systematic perturbation. In the implemented pipeline, a large number of random binary masks are generated and applied to the input image. For each masked image, the model's predicted probability for the target class (angry) is recorded. The final saliency map is obtained by aggregating the prediction scores weighted by the corresponding masks, followed by normalization for visualization.



Through this process, the RISE code operationalizes the question: "Which input regions are truly necessary for the model to maintain its prediction confidence?"

3. Conclusion

3.1. Interpretation

The **Grad-CAM** heatmap reveals a strong concentration of activation in the central upper-face region, with peak intensity around the eyes and brow area. This indicates that these regions contributed most strongly to increasing the model's confidence for the angry class. The overlay visualization further confirms that the highlighted region corresponds to facial areas associated with anger-related expressions, such as eye tension, narrowed eyes, and brow contraction. Importantly, the activation is largely confined to the face itself, with minimal emphasis on background regions. This suggests that the model's prediction is driven by relevant facial features rather than spurious correlations. The Grad-CAM output therefore provides qualitative evidence that the CNN's internal reasoning aligns with human perceptual strategies for recognizing anger(same as other emotions).

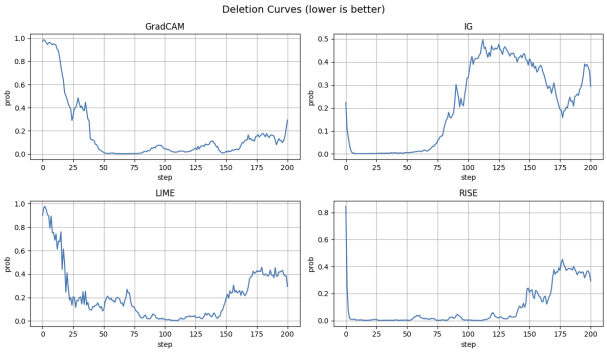
Compared to Grad-CAM, the **IG** heatmap appears more dispersed and granular, highlighting many small regions rather than a single contiguous area. This indicates that the model's prediction is influenced by distributed micro-features, such as fine intensity changes around the eyes and subtle shading variations across the face. The overlay visualization further emphasizes that the most salient pixels correspond to emotion-relevant facial features. Although some attribution noise is visible across the face, the highest-density activations remain localized around meaningful facial regions rather than the background.

The **LIME** heatmap highlights a small number of grid patches, with the most salient regions appearing around the upper-face area, including regions near the eyes. Compared to Grad-CAM and IG, the LIME explanation is noticeably less continuous, focusing on a few isolated patches rather than broad regions or dense pixel-level patterns. This suggests that the model's prediction can be strongly supported by limited but highly informative facial cues, rather than requiring the entire facial structure. The overlay visualization confirms that the most influential patches correspond to emotion-relevant facial areas, while many other facial regions contribute little to the local decision. This sparsity indicates that LIME successfully identifies a concise explanation set, even though fine-grained facial details are not explicitly represented.

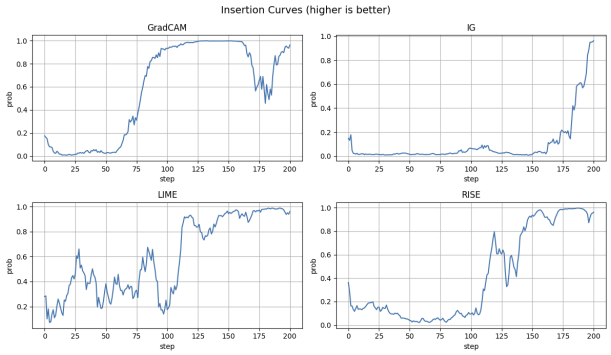
The **RISE** heatmap exhibits a scattered distribution of salient points across the facial area, with no single contiguous region dominating the explanation. At first glance, the visualization appears noisy compared to Grad-CAM or IG. However, when overlaid on the input image, the salient points remain largely confined to the facial region, with minimal emphasis on the background. This indicates that the model's prediction is disrupted primarily when facial pixels—rather than non-facial areas—are occluded. The absence of a single concentrated hotspot suggests that the model relies on distributed facial evidence, rather than a single localized feature, to sustain the emotion prediction.

3.2. Evaluation via Deletion and Insertion Curves

To quantitatively evaluate the faithfulness of each explainable AI (XAI) method, this study employs deletion and insertion curves, which are widely used metrics for assessing whether an explanation correctly identifies features that are causally important for a model's prediction. Unlike qualitative visualization, deletion and insertion curves directly measure how a model's predicted probability changes when input features identified as important are systematically removed or added. As such, these metrics provide a principled way to evaluate explanation quality beyond visual plausibility.



In this study, deletion curves are interpreted such that lower curves and smaller area under the curve (AUC) values indicate better performance, as they reflect faster destruction of the model's prediction. The results show that RISE achieves the lowest deletion AUC (0.0965), indicating that the regions identified by RISE are highly necessary for maintaining the model's confidence in the angry prediction. Grad-CAM follows with a deletion AUC of 0.1724, demonstrating that removing the regions highlighted by Grad-CAM also leads to a substantial and consistent drop in confidence. Integrated Gradients and LIME exhibit higher deletion AUC values (0.2100 and 0.2205, respectively), suggesting that removing their identified pixels degrades confidence more gradually. These findings indicate that RISE provides the most causally faithful explanation under the deletion criterion, while Grad-CAM also identifies regions that are strongly necessary for the prediction.



In the insertion experiment, the process is reversed. Starting from a baseline image with minimal information, pixels are added back in descending order of importance. The model's predicted probability is measured after each insertion step. In this case, higher curves and larger AUC values indicate better performance, as they reflect how quickly the prediction can be reconstructed using the identified features. The insertion results reveal that LIME achieves the highest insertion AUC (0.6015), followed closely by Grad-CAM (0.5719). This indicates that a relatively small number of regions identified by these methods are sufficient to rapidly restore the model's confidence. RISE shows a moderate insertion AUC (0.4171), reflecting the fact that its attributions are distributed across many pixels rather than concentrated in a small set. Integrated Gradients exhibits a notably low insertion AUC (0.0986), suggesting that although IG captures pixel-level sensitivity, these pixels alone are not sufficient to quickly reconstruct the prediction under the insertion protocol. These results highlight an important distinction between necessity and sufficiency. While Integrated Gradients identifies many sensitive pixels, these attributions are spatially dispersed and therefore less effective at rapidly rebuilding the model's confidence when added incrementally.

Taken together, the deletion and insertion curves reveal complementary strengths and limitations across XAI methods. RISE excels in identifying features that are necessary for the model's prediction, as evidenced by its strong deletion performance. In contrast, LIME and Grad-CAM perform better in terms of sufficiency, identifying compact regions that efficiently reconstruct the prediction in the insertion setting. Integrated Gradients provides fine-grained sensitivity information but shows weaker performance under insertion due to its distributed attribution pattern.

3.3. Conclusion

Overall, the results support the hypothesis proposed in this study—that CNN-based FER models rely on psychologically meaningful facial cues, particularly in the eye and upper-face regions, when making emotion predictions. By combining qualitative visualization with quantitative faithfulness evaluation, this work contributes to a more transparent understanding of deep learning–based affective computing systems.

Future work may extend this analysis by exploring how explainable facial emotion recognition models can be applied in real-world domains where interpretability is essential. In human–computer interaction, XAI-enhanced FER systems could enable adaptive interfaces that respond to users' emotional states while allowing designers to understand and validate the system's behavior. In mental health and well-being applications, interpretable emotion recognition may support emotion-aware assessment tools, where transparency is critical for ethical use and clinical trust. Additionally, in safety-critical or socially sensitive contexts such as surveillance, driver monitoring, and customer-facing AI services, explainable FER can help mitigate bias, support accountability, and facilitate human oversight. By extending XAI analysis toward these application domains, future research can strengthen the practical relevance of interpretable emotion recognition and promote responsible deployment of emotion-aware AI systems.

References

[1] Hannes Eisenbarth and Georg W. Alpers. Eye movements and facial expression recognition: Same information, same time? *Journal of Nonverbal Behavior*, 35(3):275–286, 2011. 1

[2] Ian J. Goodfellow, Dumitru Erhan, Pierre Luc Carrier, Aaron Courville, Mehdi Mirza, Ben Hamner, Will Cukierski, Yichuan Tang, David Thaler, Dong-Hyun Lee, et al. Challenges in representation learning: A report on three machine learning contests. In *Proceedings of the International Conference on Neural Information Processing (ICONIP)*, pages 117–124. Springer, 2013. 1